



The author, Antonio Espingardeiro, has created the P37 S65 robot that has the ability to remind elderly people to take their medication and exercise, and it can even tell jokes

cles. Similarly, the use of unmanned aerial vehicles (UAVs) for agriculture checks, treatment and mapping will become a reality in the future. This is likely to be translated into a reduction of farming costs, and inevitably into a reduction of food prices.

Materials research is likely to offer alternatives in robotics production and related areas. In the future, robots could become cheaply-built, disposable and recyclable in numerous applications. On the same spectrum, robots could become important tools for collecting garbage and recycling. Cleaning parks, oceans or other areas could become a reality for the use of robotics and AI. Such features could

have a positive impact on the environment.

In terms of space robotics, we could also see some increase in the use of autonomy in space explorations. Autonomy could be translated in terms of orbital robotics when assisting astronauts, for example tossing satellites, opening or closing valves, or simply cleaning certain equipment.

On another spectrum, one of the biggest challenges with planetary rovers (ROVs) deals with delays in communication between the Earth and robots. Meanwhile, AI algorithms are being developed for giving the robot enough autonomy to explore certain areas while humans try to regain control. Such challenge is highly important in terms of negotiating trajectories

to avoid obstacles, maintain the robot's integrity, save energy and possibly regain communications with the Earth.

Another aspect to bear in mind is that ROVs can act as scientific advisors (expert systems) for human operators back on the Earth (for example, for sensing things that would be biologically impossible to detect) in dangerous environments.

By now it is perceptible that when performing research into autonomous navigation, we are not only talking about autonomous driving vehicles, but indeed pushing the boundaries of robotics and AI to other areas of human life. ●



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